higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

COMMUNICATION ELECTRONICS N4

(8080224)

15 November 2019 (X-paper)
09:00–12:00

This question paper consists of 7 pages and 1 formula sheet.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
COMMUNICATION ELECTRONICS N4
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a new page.
5. Use only a blue or black pen.
6. Write neatly and legibly.

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QUESTION 1

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 An antenna in any receiver converts audio to sound.

1.2 Ultraviolet radiation from the sun is the primary cause of ionisation.

1.3 A microphone converts audio signals into electromagnetic waves.

1.4 In a klystron amplifier, the function of the catcher cavity is to emit a stream of electrons.

1.5 The impedance of a series circuit at resonance is a minimum.

1.6 An AM signal with 110% modulation is over-modulated.

1.7 A multicavity magnetron is an example of a modulator in radio communication.

1.8 The construction of a rhombic antenna is very cheap.

1.9 Radio waves travel at a speed of 3×10^6 metres per second.

1.10 A radio antenna is directional.

(10 × 1) **[10]**

QUESTION 2

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–O) next to the question number (2.1–2.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
2.1	The law that states that the sum of the EMF in any closed circuit is equal to the sum of the potential difference in that circuit	A	cut-off frequency
2.2	The theorem that has as one of its strategies the elimination of all but one source of power within a network at a time	B	independent side band
2.3	The resonance circuit that has as one of its conditions, reducing the circuit current to its minimum	C	pass band
2.4	The point in the response curve of the filter where attenuation changes from zero to a large value	D	series-resonant circuit
2.5	The low-pass filter is connected in series with a high-pass filter, with the cut-off frequency of the low-pass filter higher than that of the high-pass filter	E	differentiator
2.6	The RC circuit with the values of R and C chosen in such a way that 5 RC is much less than t	F	Kirchoff's second law
2.7	The transmission in which the upper and lower side bands, obtained from two separate modulation circuits using the same carrier frequency, are combined	G	demodulation
2.8	The process of returning the modulated data signals to their original form	H	parallel resonant circuit
2.9	An amplifier which is widely used in microdevices for radio-wave propagation	I	travelling-wave tube
2.10	An amplifier which is widely used for continuous-wave or pulsed-radar transmitters	J	resonance
		K	multicavity klystron
		L	modulation
		M	superposition
		N	reflex klystron
		O	stop band

(10 × 1) **[10]**

QUESTION 3: NETWORK THEOREMS

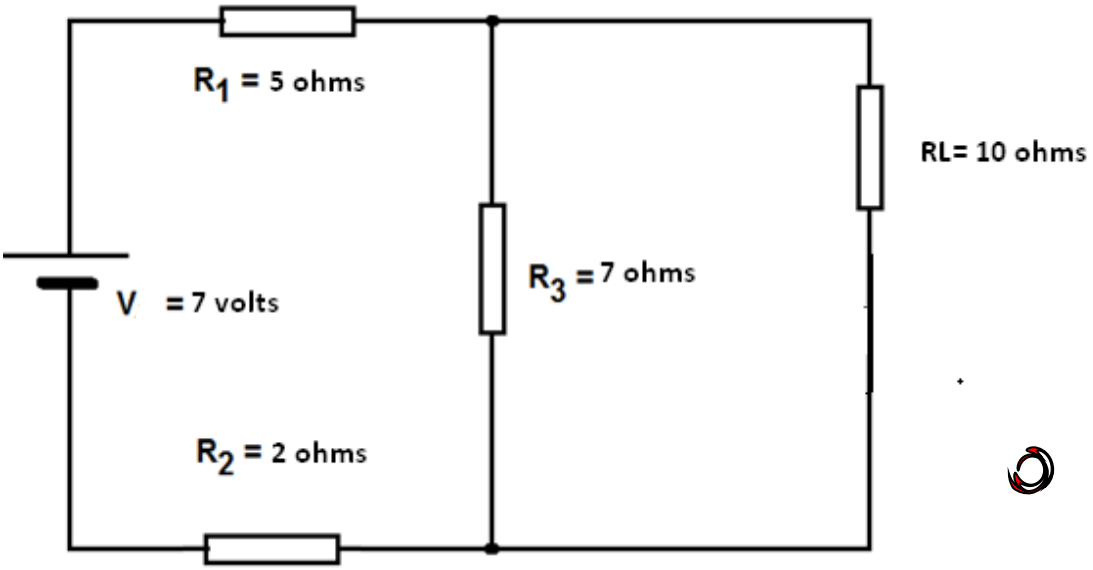
3.1 Define the following:

3.1.1 Kirchhoff's voltage law

3.1.2 Norton's theorem

(2 × 2) (4)

3.2 Use Norton's theorem to find Norton's equivalent circuit of the network shown below. Draw the equivalent network.



(6)
[10]

QUESTION 4: ALTERNATING CURRENT THEORY

A series circuit consists of a coil with a DC resistance of 15 Ω and an inductance of 10 mH in series and a variable capacitor.

4.1 Prove, by means of relevant formulae, that $C = \frac{1}{4\pi^2 f^2 L}$ at resonance. (3)

4.2 Determine the capacitance at resonance if the resonance frequency is 85 kHz. (1)

4.3 Determine the following:

4.3.1 The bandwidth

4.3.2 The Q-factor

4.3.3 The impedance of the circuit at resonance

(3 × 1) (3)
[7]

QUESTION 5: ATTENUATORS AND FILTERS

5.1 Make a neat circuit diagram of each of the following attenuators:

5.1.1 O-type

5.1.2 H-type

(2 × 1) (2)

5.2 Make a neat diagram of each of the following elements:

5.2.1 Band-pass filter (1)

5.2.2 Attenuation versus frequency graph for the filter mentioned in QUESTION 5.2.1 (2)

[5]

QUESTION 6: MODULATION

6.1 Define the following:

6.1.1 Frequency modulation

6.1.2 Modulation index

(2 × 2) (4)

6.2 Explain the following processes:

6.2.1 Modulation process

6.2.2 Radio communication process

(2 × 2) (4)

[8]

QUESTION 7: DEMODULATION

7.1 Draw a fully labelled circuit diagram of an AM half-wave detector. (3)

7.2 Explain the operation of the AM half-wave detector mentioned in QUESTION 7.1. (7)

[10]

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QUESTION 8: COMMUNICATION SYSTEMS

8.1 Draw a block diagram of an FM superheterodyne receiver. (5)

8.2 Explain the function of each stage of the receiver mentioned in QUESTION 8.1. (10)

[15]

QUESTION 9: RADIO WAVE PROPAGATION

9.1 Define the following:

9.1.1 Silent zone

9.1.2 Critical frequency

9.1.3 Polarisation

(3 × 3) (9)

9.2 Make a neat, fully labelled sketch of a travelling-wave tube. (3)

9.3 Explain the operation of the travelling-wave tube mentioned in QUESTION 9.2. (7)

9.4 Explain the difference between *directional antenna* and *omnidirectional antenna*. (2)

9.5 Make a neat, fully labelled sketch of each of the following:

9.5.1 A rhombic antenna

9.5.2 A folded dipole antenna

(2 × 2) (4)

[25]

TOTAL: 100

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COMMUNICATION-ELECTRONICS N4

FORMULA SHEET

1. $I = \frac{V}{R}$

2. $X_L = 2\pi fL$

3. $X_C = \frac{1}{2\pi fc}$

4. $Z = \sqrt{R^2 + (X_L - X_C)^2}$

5. $I = \frac{V}{Z}$

6. $V_R = I \,.\, R$

   $V_L = I \,.\, X_L$

   $V_C = I \,.\, X_C$

7. $Cos\phi = \frac{R}{Z}$

8. $I = \sqrt{I_R + I_C^2}$

9. $\theta = Tan^{-1}\frac{I_C}{I_R}$

10. $I = \sqrt{I_R^2 + I_L^2}$

11. $I = \sqrt{I_R^2 \ + (I_L - L_C)^2}$

12. $\theta = Tan^{-1}\frac{(I_L - I_C)}{I_R}$

13. $f_r = \frac{1}{2\pi\sqrt{LC}}$

14. $C = \frac{1}{4 \,.\, \pi^2 \,.\, f_r^2 \,.\, L}$

15. $L = \frac{1}{4 \,.\, \pi^2 \,.\, f_r^2 \,.\, C}$

16, $Q = \frac{X_L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{f_R}{band\ width/bandwydte}$

17. $V_c = V \times Q$ *and/en* $V_L = V \times Q$

18. $Z = \frac{L}{CR}$

19. $I_c = I \times Q$

20. $I_L = I \times Q$

21. $f = \frac{\nu}{\lambda}$

22. $B - W = \frac{R}{2\pi L}$

23. $V_T = \sqrt{V_R{}^2 + (V_C - V_L)^2}$

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